VIBHOR SHESHRAO PATIL

Full Stack Engineer • Distributed Systems & FinTech Architecture

Nagpur, India • Open to Remote (India-wide) | +91 8459005237 | vibhorpatil808@gmail.com | linkedin.com/in/vibhor-patil | github.com/IamVibhor

TECHNICAL SKILLS

Languages & Frameworks:	TypeScript, Node.js, NestJS, JavaScript (ES6+), PHP, Laravel, Vue.js, Express.js, Tailwind CSS
Databases & Caching:	PostgreSQL, MySQL, Redis (Distributed Locks, Caching), Prisma ORM, Sequelize
Architecture & Protocols:	Finite State Machines, Idempotency-Key Design, Double-Entry Accounting, Liveness KYC, Aadhaar QR / PAN / Voter APIs, Surepass Aadhaar E-Sign, RBAC, Microservices
DevOps & Cloud:	Docker, Docker Compose, AWS (S3, EC2), RESTful APIs, Swagger/OpenAPI, Jest, Git, CI/CD Pipelines

WORK EXPERIENCE

YELLOW METAL FINTECH PVT. LTD
Remote, India

Full Stack Engineer (NestJS, TypeScript, PostgreSQL, Prisma, Vue.js, Docker)
03/2025 – Present

- **13-Stage Lending Engine:** Architected and engineered the core automated digital loan origination pipeline (LOS) spanning a high-throughput **NestJS backend** and a responsive **Vue.js underwriting admin portal**.
- **Turnaround Time (TAT) Optimization:** Compressed loan origination turnaround from **48 hours to 12.4 minutes (-95.7%)** by automating applicant ingestion, risk qualification, credit bureau pulls, and bank disbursements.
- **Automated KYC & Liveness Engine:** Engineered a multi-tier KYC verification pipeline integrating live facial biometric liveness checks, PAN validation, Aadhaar Secure QR decoding, and Voter ID verification; enforced immutable database-level locks post-verification to prevent document alteration fraud.
- **FinTech & Regulatory Integrations:** Integrated Surepass eSign gateway for Aadhaar OTP-based digital loan agreement execution; built automated credit bureau evaluation pipelines with CIF customer profiling, risk scoring, and statutory audit logging.
- **Idempotent Disbursement Rails:** Built resilient core banking disbursement workflows over IMPS/NEFT with **atomic Redis Idempotency-Key caching** and a double-entry ledger to guarantee **zero double-payouts** during gateway timeouts.
- **High-Performance Migration:** Spearheaded legacy monolith migration to modular NestJS + Prisma ORM microservices, optimizing query indexing and reducing **p99 API latencies by 38%** (<65ms response times).

EMT TIMES PVT. LTD
Nagpur, India

Backend Engineer (PHP, Laravel, MySQL, Redis, REST APIs)
07/2024 – 02/2025

- Scaled high-throughput REST APIs and monetization services supporting **50,000+ daily active users** with high availability.
- Engineered multi-tier Redis caching strategies and optimized relational queries, cutting database CPU utilization by **25%** under peak surges.
- Developed fault-tolerant background queue workers for asynchronous SMS, push notifications, and automatic webhook retries.

ICEICO TECHNOLOGIES PVT. LTD
Nagpur, India

Software Engineer (PHP, Laravel, Vue.js, MySQL, Payment Gateways)
09/2022 – 06/2024

- Architected enterprise CRM workflows featuring automated lead progression, deal stage tracking, and dynamic RBAC access controls.
- Integrated Razorpay payment processing and automated settlement workflows with cryptographic HMAC-SHA256 webhook signature reconciliation.
- Built modular, component-driven UI dashboards in Vue.js and Tailwind CSS, standardizing reusable frontend design patterns.

KEY ARCHITECTURAL HIGHLIGHTS

Automated FinTech Loan Origination System (LOS) — NestJS • PostgreSQL • Prisma • Redis • Docker

- Engineered a deterministic 13-stage state machine with strict transition guards, role-gated token permissions across Consumer, Loan Manager, and Admin modules, and audit trails ensuring strict compliance with RBI digital lending guidelines.

Distributed Payment & Ledger Engine — Node.js • Redis Idempotency • Double-Entry Ledger

- Engineered a resilient payment processing architecture featuring atomic Idempotency-Key caching (zero duplicate charges), HMAC-SHA256 webhook security, and a balanced double-entry accounting ledger ensuring mathematical parity across financial transactions.

EDUCATION

Master of Science in Computer Science (M.Sc CS) • Advanced Software Engineering, Cloud & System Design — 08/2023 – 06/2025

Bachelor of Science in Information Technology (B.Sc IT) • Core Computer Science, Algorithms, Database Systems — 05/2019 – 05/2022